Ch. 11 — P02 (A) Antithetic Variates

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The antithetic variates method pairs each run with a "mirror" run using $1-U$ in place of U for each uniform deviate. For a monotone increasing estimator $g(U)$, this induces negative correlation between paired runs and reduces variance:

$$\mathrm{Var}\left(\frac{g(U)+g(1-U)}{2}\right) = \frac{\mathrm{Var}[g(U)] + \mathrm{Cov}[g(U),\, g(1-U)]}{2}$$

Tasks

1. **Analytical example:** Let $U \sim \mathrm{Uniform}(0,1)$ and $g(U) = U^2$. Compute:
   (a) $E[g(U)] = E[U^2]$
   (b) $\mathrm{Var}[g(U)]$
   (c) $\mathrm{Cov}[g(U),\, g(1-U)]$
   (d) $\mathrm{Var}[(g(U)+g(1-U))/2]$

   What is the variance reduction ratio compared to using n independent runs?
2. **Exponential service times:** For an M/M/1 queue, service times use the inverse CDF: $S = -\frac{1}{\mu}\ln(U)$. Show that $S' = -\frac{1}{\mu}\ln(1-U)$ is also Exponential(μ). Does this guarantee negative correlation between $W_q(S)$ and $W_q(S')$? Under what conditions on the queueing model is the answer yes?
3. **Coding:** Implement an M/M/1 simulation ($\lambda = 0.7$, $\mu = 1.0$, 1000 customers) that:
   (a) Runs $n = 20$ independent replications
   (b) Runs $n = 10$ antithetic pairs (20 runs total, paired)

   Report the standard error of $\hat{W}_q$ under each approach. Compute the variance reduction ratio.
4. **Limitation:** Antithetic variates require g to be monotone in the input uniforms to guarantee a variance reduction. Describe a queueing scenario where this monotonicity breaks down. *(Hint: consider priority queues or finite buffers.)*